| Program : **Diploma in Biomedical Engineering** | |
|---|---|
| Course Code : **5242** | Course Title: **Hospital Systems** |
| Semester : **5** | Credits: **4** |
| Course Category: **Program Core** | |
| Periods per week: **4 (L:3, T:1, P:0)** | Periods per semester: **60** |

**Course Objectives:**

- To impart thorough knowledge about Hospital Organization and its Architectural design requirements.
- To lay foundation on the major systems in hospitals like MGPS, CSSD, hospital communication, and telemedicine.
- To enhance the knowledge on health regulatory requirement and equipment maintenance management.

**Course Prerequisites:**

| Topic | Course code | Course name | Semester |
|---|---|---|---|
| Basic idea about electrical and electronic circuits | | Fundamentals of electrical and electronics engineering | 2 |

**Course Outcomes:**

On completion of the course, the student will be able to:

| COn | Description | Duration (Hours) | Cognitive Level |
|---|---|---|---|
| CO1 | Illustrate the design of major departments in hospitals | 18 | Understanding |
| CO2 | Describe the Electrical Power Distribution System, Medical Gas Pipeline Systems (MGPS), Central Sterile Supply Department (CSSD) in hospitals | 18 | Understanding |
| CO3 | Explain the basic telecommunication systems and biotelemetry systems in hospitals. | 11 | Understanding |
| CO4 | Describe the regulatory requirements and health care codes | 11 | Understanding |
| | Series Test | 2 | |

**CO – PO Mapping:**

| Course Outcomes | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 |
|---|---|---|---|---|---|---|---|
| **CO1** | 2 | | | | | | |
| **CO2** | 2 | | | | | | |
| **CO3** | 2 | | | | | | |
| **CO4** | 2 | | | | | | |

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

**Course Outline:**

<table>
<tr><th>Module Outcomes</th><th>Description</th><th>Duration (Hours)</th><th>Cognitive Level</th></tr>
<tr><td>CO1</td><td colspan="3">Illustrate the design of major Departments in hospitals</td></tr>
<tr><td>M1.01</td><td>Describe the types of healthcare organizations. Illustrate and detail the hospital organization chart and role of Biomedical Engineer</td><td>6</td><td>Understanding</td></tr>
<tr><td>M1.02</td><td>Summarize the Modern hospital architecture and functional planning of hospital</td><td>2</td><td>Understanding</td></tr>
<tr><td>M1.03</td><td>Distinguish the design of major departments in hospital</td><td>10</td><td>Understanding</td></tr>
<tr><td colspan="4">Contents:<br>Hospital Organization, Hospital Planning, Design of major Departments<br>Definition of Bio-engineering, bio-medical Engineering, Clinical Engineering and Hospital Engineering. Describe the types of healthcare Organization. Explain the hospital organisation chart. Duties and responsibilities of Biomedical engineer in a hospital.<br>Modern Hospital Architecture and functional Planning of a Hospital - Design of Operation theatre, patient wards, ICU, Emergency department, Radiology room</td></tr>
<tr><td>CO2</td><td colspan="3">Describe the Electrical Power Distribution System, Medical Gas Pipeline Systems (MGPS), Central Sterile Supply Department (CSSD) in hospitals.</td></tr>
<tr><td>M1.01</td><td>Summarize the Electrical Power Distribution System in Hospital, Electrical Safety requirement & Isolated Power System.</td><td>5</td><td>Understanding</td></tr>
<tr><td>M2.01</td><td>Explain Medical Gas Pipeline Systems(MGPS) for Centralized gas supply, liquid oxygen plant and vacuum supply system in hospitals with suitable schematics</td><td>8</td><td>Understanding</td></tr>
<tr><td>M2.02</td><td>Summarize the need for CSSD department, Sterile Supply Cycle and different types of sterilizer used in hospitals</td><td>5</td><td>Understanding</td></tr>
<tr><td></td><td>Series Test – I</td><td>1</td><td></td></tr>
</table>

**Contents:**

**Electrical Power Distribution System, Medical Gas Pipeline System (MGPS), Central Sterile Supply Department(CSSD) in hospitals**

Electrical power distribution systems in Hospitals - Schematic of power distribution system, different types of power sources, distribution circuits - Importance of uninterrupted power supply and voltage stabilizers in Hospital

Electrical safety - Basic electric safety measures in hospital, Physiological effect of electrical currents, micro shock and macro shock. Isolated Power System-Isolation Transformer

Medical Gas Pipeline Systems (MGPS) for Centralized gas supply - need - safety requirements - schematic - working, color coding for pipelines. Centralized vacuum supply system - need - schematic - working. Liquid oxygen plant - schematic and working principle

Central Sterile Supply Department(CSSD) - need, explain Sterile supply cycle, Sterilization techniques - Types of sterilizer - Single and double chamber autoclave - Steam sterilizer - Ethylene oxide Gas sterilizer (Schematic and working)

| **CO3** | **Explain the basic telecommunication systems and biotelemetry systems in hospitals.** | | |
|---|---|---|---|
| M3.01 | Elaborate the elements of a communication system and modulation techniques | 3 | Understanding |
| M3.02 | Explain the transmitter and receiver section of a communication system | 3 | Understanding |
| M3.03 | Describe the basic system of biotelemetry | 2 | Understanding |
| M3.04 | Illustrate the telemetric system components | 3 | Understanding |

**Contents:**

**Communication systems and Biotelemetry**

Communication systems - basic concepts - block diagram- modulation- definition - analog modulation techniques - AM, FM, PM - digital modulation techniques - ASK, FSK, PSK (concepts and waveforms ) - transmitter section - receiver section - block diagram

Biotelemetry - definition - physiological parameters - system components - single channel telemetry - multi channel telemetry (block diagram and working principle)

| **CO4** | **Describe the regulatory requirements and health care codes** | | |
|---|---|---|---|
| M4.01 | Explain the important national and international accreditation organization for Hospital | 4 | Understanding. |
| M4.02 | Familiarize with ISO certification process of hospitals | 2 | Understanding. |

| | | | |
|---|---|---|---|
| M4.03 | Explain the equipment maintenance in hospitals | 5 | Understanding. |
| | Series Test – II | 1 | |

**Contents:**

**Hospital Accreditation & Equipment Maintenance Management**

Accreditation definition - National and International agencies - Explain Joint commission International, NABH, ISO

Maintenance of Equipment - Difference between Preventive maintenance & Breakdown Maintenance. Maintenance types - In house & contract - Contract Management - AMC. Testing and Calibration of Medical Equipment - Definition, need, Procedure

**Text / Reference:**

| T/R | Book Title/Author |
|---|---|
| R1 | G.D. Kunders - *Hospitals: Facilities Planning and Management,* Tenth edition, Tata McGraw Hill-India, 2008 |
| R2 | John G. Webster - *Encyclopedia of Medical Devices and Instrumentation(*6 Volume set) *,* Wiley,7 April 2006 |
| R3 | Step by Step Hospital Designing and Planning, Dr Malhotra Series,JAYPEE |
| R4 | R S Khandpur – *TELEMEDICINE -Technology and applications (mHealth, TeleHealth and eHealth),* PHI learning private LTD. Delhi, First edition, 2017 |

**Online Resources:**

| Sl.No | Website Link |
|---|---|
| 1 | eBook (Fallout protection for hospitals, United states, public health services) |
| 2 | https://www.tutorialspoint.com |
| 3 | https://nptel.ac.in/courses |